

Jessica Amihere
57592028
Cohort B

```
MariaDB [(none)]> create database EmployeeManagementDB;  
Query OK, 1 row affected (0.010 sec)
```

```
MariaDB [(none)]> use EmployeeManagementDB;  
Database changed
```

```
MariaDB [EmployeeManagementDB]> create table employee(empNo int not null, fName  
varchar(100) not null, lName varchar(100), address varchar(100), dob date, sex  
char, position varchar(100), deptNo int, primary key (empNo));  
Query OK, 0 rows affected (0.086 sec)
```

```
MariaDB [EmployeeManagementDB]> create table department (deptNo int not null,  
deptName varchar(100) not null, mgrEmpNo int unique, foreign key(mgrEmpNo)  
references employee(empNo), primary key(deptNo));  
Query OK, 0 rows affected (0.036 sec)
```

```
MariaDB [EmployeeManagementDB]> alter table employee add foreign key(deptNo)  
references department(deptNo);  
Query OK, 0 rows affected (0.067 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [EmployeeManagementDB]> create table project(projNo int not null, projName  
varchar(100), deptNo int not null, foreign key(deptNo) references  
department(deptNo), primary key (projNo));  
Query OK, 0 rows affected (0.034 sec)
```

```
MariaDB [EmployeeManagementDB]> create table worksOn(empNo int not null, projNo int  
not null, dateWorked date, hoursWorked decimal, foreign key(empNo) references  
employee(empNo), foreign key(projNo) references project(projNo), primary key(empNo,  
ProjNo, dateWorked));  
Query OK, 0 rows affected (0.037 sec)
```

```
MariaDB [EmployeeManagementDB]> insert into department values  
-> (1, 'IT', 1),  
-> (2, 'Planning', 4),  
-> (3, 'Finance', 6),  
-> (4, 'HR', 2),  
-> (5, 'Marketing', 5),  
-> (6, 'Operations', 3),  
-> (7, 'Research', 7),  
-> (8, 'Legal', 8),  
-> (9, 'Sales', 9),  
-> (10, 'Support', 10);  
Query OK, 10 rows affected (0.020 sec)  
Records: 10 Duplicates: 0 Warnings: 0
```

```

MariaDB [EmployeeManagementDB]> insert into employee values
-> (1, 'Alice', 'Smith', '12 Oak St', '1985-03-14', 'F', 'Manager', 1),
-> (2, 'Bob', 'Jones', '34 Pine Ave', '1990-07-22', 'M', 'Analyst', 1),
-> (3, 'Carol', 'White', '56 Elm Rd', '1988-11-05', 'F', 'Software Developer',
2),
-> (4, 'David', 'Brown', '78 Maple Dr', '1992-01-30', 'M', 'Team Leader', 2),
-> (5, 'Emma', 'Taylor', '90 Birch Ln', '1995-06-18', 'F', 'Analyst', 3),
-> (6, 'Frank', 'Davis', '11 Cedar Ct', '1983-09-09', 'M', 'Manager', 3),
-> (7, 'Grace', 'Wilson', '22 Walnut Way', '1991-04-25', 'F', 'Software
Developer', 1),
-> (8, 'Henry', 'Moore', '33 Ash Blvd', '1987-12-12', 'M', 'Analyst', 2),
-> (9, 'Isla', 'Clark', '44 Poplar Pl', '1993-08-07', 'F', 'Team Leader', 3),
-> (10, 'Jack', 'Lewis', '55 Spruce St', '1996-02-14', 'M', 'Software
Developer', 1);
Query OK, 10 rows affected (0.003 sec)
Records: 10  Duplicates: 0  Warnings: 0

```

```

MariaDB [EmployeeManagementDB]> insert into project values
-> (1, 'SCCS', 1),
-> (2, 'Payroll', 2),
-> (3, 'BudgetApp', 3),
-> (4, 'HRPortal', 4),
-> (5, 'AdCampaign', 5),
-> (6, 'OpsTracker', 6),
-> (7, 'LabSystem', 7),
-> (8, 'ComplianceTool', 8),
-> (9, 'SalesDash', 9),
-> (10, 'HelpDesk', 10);
Query OK, 10 rows affected (0.022 sec)
Records: 10  Duplicates: 0  Warnings: 0

```

```

MariaDB [EmployeeManagementDB]> insert into worksOn values
-> (1, 1, '2026-01-05', 8),
-> (2, 1, '2026-01-06', 6),
-> (3, 2, '2026-01-07', 7),
-> (4, 2, '2026-01-08', 5),
-> (5, 3, '2026-01-09', 8),
-> (6, 3, '2026-01-10', 4),
-> (7, 4, '2026-01-11', 6),
-> (8, 5, '2026-01-12', 7),
-> (9, 6, '2026-01-13', 8),
-> (10, 7, '2026-01-14', 5);
Query OK, 10 rows affected (0.023 sec)
Records: 10  Duplicates: 0  Warnings: 0

```

```

MariaDB [EmployeeManagementDB]>
MariaDB [EmployeeManagementDB]> set foreign_key_checks = 1;
Query OK, 0 rows affected (0.000 sec)

```

```
a)MariaDB [EmployeeManagementDB]> DELIMITER //
MariaDB [EmployeeManagementDB]> create trigger s_trigger
-> before insert on employee
-> for each row
-> begin
-> if new.sex not in ('F', 'M')
-> then
-> signal sqlstate '45000'
-> set message_text = 'Invalid sex entered';
-> end if;
-> end//
```

Query OK, 0 rows affected (0.040 sec)

```
b)MariaDB [EmployeeManagementDB]> create trigger p_trigger
-> before insert on employee
-> for each row
-> begin
-> if new.position not in ('Manager', 'Team leader', 'Analyst', 'Software
Developer')
-> then
-> signal sqlstate '45000'
-> set message_text = 'Invalid position entered';
-> end if;
-> end//
```

Query OK, 0 rows affected (0.043 sec)

```
c)MariaDB [EmployeeManagementDB]> create trigger h_trigger
-> before insert on worksOn
-> for each row
-> begin
-> if new.hoursWorked not between 0 and 40
-> then
-> signal sqlstate '45000'
-> set message_text = 'Hours not within range';
-> end if;
-> end//
```

Query OK, 0 rows affected (0.033 sec)

```
alter table worksOn modify hoursWorked int//
```

Query OK, 10 rows affected (0.080 sec)

Records: 10 Duplicates: 0 Warnings: 0

7.25

```
MariaDB [EmployeeManagementDB]> create view projects_femaleManagers as select
e.fname, e.lname, p.projName, p.projNo from employee e, project p, department d
where p.deptNo = d.deptNo and d.deptNo = e.deptNo and e.sex = 'F' group by
p.projNo;
```

Query OK, 0 rows affected (0.054 sec)

```
MariaDB [EmployeeManagementDB]> select * from projects_femaleManagers;
```

fname	lname	projName	projNo
Alice	Smith	SCCS	1
Carol	White	Payroll	2
Emma	Taylor	BudgetApp	3

3 rows in set (0.029 sec)

7.26

MariaDB [EmployeeManagementDB]> create view project_managers as select e.empNo, e.fname, e.lname, p.projName, w.hoursWorkded from employee e, project p, department d, worksOn w where p.deptNo= d.deptNo and d.deptNo= e.deptNo and w.projNo = p.projNo;

Query OK, 0 rows affected (0.028 sec)

MariaDB [EmployeeManagementDB]> select * from project_managers;

empNo	fname	lname	projName	hoursWorkded
1	Alice	Smith	SCCS	8
1	Alice	Smith	SCCS	6
2	Bob	Jones	SCCS	8
2	Bob	Jones	SCCS	6
3	Carol	White	Payroll	7
3	Carol	White	Payroll	5
4	David	Brown	Payroll	7
4	David	Brown	Payroll	5
5	Emma	Taylor	BudgetApp	8
5	Emma	Taylor	BudgetApp	4
6	Frank	Davis	BudgetApp	8
6	Frank	Davis	BudgetApp	4
7	Grace	Wilson	SCCS	8
7	Grace	Wilson	SCCS	6
8	Henry	Moore	Payroll	7
8	Henry	Moore	Payroll	5
9	Isla	Clark	BudgetApp	8
9	Isla	Clark	BudgetApp	4
10	Jack	Lewis	SCCS	8
10	Jack	Lewis	SCCS	6

20 rows in set (0.003 sec)

7.27

MariaDB [EmployeeManagementDB]> create view EmpProject(empNo, projNo, totalHours) as select w.empNo, w.projNo, sum(hoursWorkded) from employee e, project p, worksOn w where e.empNo = w.empNo and p.projNo = w.projNo group by w.empNo, w.projNo;

Query OK, 0 rows affected (0.030 sec)

a)MariaDB [EmployeeManagementDB]> select * from EmpProject;

empNo	projNo	totalHours
1	1	8
2	1	6
3	2	7
4	2	5
5	3	8
6	3	4
7	4	6
8	5	7
9	6	8
10	7	5

10 rows in set (0.003 sec)

b)MariaDB [EmployeeManagementDB]> select projNo from EmpProject where projNo='SCCS';
Empty set, 1 warning (0.023 sec)

c)MariaDB [EmployeeManagementDB]> select count(projNo) from EmpProject where empNo = 1;

count(projNo)
1

1 row in set (0.001 sec)

d)MariaDB [EmployeeManagementDB]> select empNo, totalHours from empProject group by empNo;

empNo	totalHours
1	8
2	6
3	7
4	5
5	8
6	4
7	6
8	7
9	8
10	5

10 rows in set (0.001 sec)

7.28

When a new row is inserted, the view can be maintained directly without

checking the Part table. If the new row has a partCost > 1000 and its partNo is not already in the view, then it is added, otherwise, no changes are necessary. For delete and update, I would check the base table first doing anything to the view because the same partNo may appear in other rows, due to the distinct used.